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GAME BALL WITH VARIABLE LOBULAR SHAPE

Abstract

A variable loby game ball has an outer surface that includes a first pole, a second pole opposite the first pole and outer lobes. The outer lobes extend between the first pole and the second pole. Recessed channels that vary in at least one of width, depth and cross-sectional shape extend along consecutive lobes.

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Background/Summary

BACKGROUND

[0001] Game balls sometimes include an outer lobular surface formed by a series of lobes

extending between opposite poles of the ball. Recessed channels separate the lobes and facilitate gripping of the game ball. One example of such a game ball is a basketball.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0002] FIG. 1 is a sectional view along a first plane through an equator of an example game ball.
- [0003] FIG. 2 is a sectional view along a second plane between the first plane and a pole of the example game ball of FIG. 1.
- [0004] FIG. 3 is a sectional view along a first plane through an equator of an example game ball.
- [0005] FIG. 4 to is a sectional view along a second plane between the first plane and a pole of the example game ball of FIG. 3.
- [0006] FIG. 5 is a perspective view of an example game ball comprising a variable lobular outer surface.
- [0007] FIG. 6 is a perspective view of an example game ball comprising a variable lobular outer surface.
- [0008] FIG. 7 is a perspective view of an example game ball having a variable lobular outer surface.
- [0009] FIG. 8 is a top view of the game ball of Figure, the bottom being a mirror image of FIG. 8.
- [0010] FIG. 9 is a front view of the example game ball of FIG. 7.
- [0011] FIG. 10 is a side view of the example game ball of FIG. 7.
- [0012] FIG. 11 is an enlarged view of a portion of the game ball of FIG. 9 proximate to an equator of the game ball.
- [0013] FIG. 12 is an enlarged view of portion of the game ball of FIG. 9 proximate to a pole of the game ball.
- [0014] FIG. 13 is an enlarged view of a portion of the game ball of FIG. 9 proximate to the equator of the game ball.
- [0015] FIG. 14 is a sectional view of the enlarged portion of FIG. 13.
- [0016] FIG. 15 is an enlarged view of a portion of the game ball of FIG. 15 proximate to a pole of the game ball.
- [0017] FIG. 16 is a sectional view of the portion of FIG. 15.
- [0018] FIG. 17 is a sectional view of the game ball of FIG. 9 at a location where transitioning sides of a recessed channel on an outer surface of the game ball have their shallowest depth.
- [0019] FIG. 18 is a sectional view of the game ball of FIG. 17 at a location along the recessed channel where the transitioning sides of the recessed channel begin to transition to a deeper depth.
- [0020] FIG. 19 is a sectional view of the game ball of FIG. 17 at a location along the recessed channel where the transitioning sides of the recessed channel have their deepest depth.
- [0021] FIG. 20 is a perspective view of an example game ball having a variable lobular outer surface.
- [0022] FIG. 21 is a perspective view of an example game ball having a variable lobular outer surface.
- [0023] FIG. 22 is a perspective view of an example game ball having a variable lobular outer surface.
- [0024] FIG. 23 is a sectional view of portions of an example game ball having a variable lobular outer surface.
- [0025] FIG. 24 is a sectional view of portions of an example game ball having a variable lobular outer surface.
- [0026] FIG. 25 is a sectional view of portions of an example game ball having a variable lobular outer surface.

[0027] Throughout the drawings, identical reference numbers designate similar, but not necessarily identical, elements. The figures are not necessarily to scale, and the size of some parts may be exaggerated to more clearly illustrate the example shown. Moreover, the drawings provide examples and/or implementations consistent with the description; however, the description is not limited to the examples and/or implementations provided in the drawings.

DETAILED DESCRIPTION OF EXAMPLES

[0028] Disclosed are example game balls that have outer surfaces configured to facilitate enhanced gripping of the game ball. The outer surfaces of the example game balls comprise a first pole and a second pole opposite the first pole. The outer surfaces comprise outer circumferential portions or outer lobes that extend between the first pole and the second pole. In some implementations, the outer lobes extend from the first pole to the second pole. Recessed channels that vary in at least one of width, depth and cross-sectional shape extend along consecutive lobes.

[0029] In some implementations, each of the recessed channels comprises a floor and a pair of transitioning sides on opposite sides of the floor. The transitioning sides extend from proximate the floor outwardly to the outermost circumferential surface of the game ball. In some implementations, a surface of the floor is outwardly concave in a direction perpendicular to a centerline of the recessed channel and convex in a direction parallel to the centerline of the recessed channel. In some implementations, the surface of the floor is flat in a direction perpendicular to a centerline of the recessed channel and is convex in a direction parallel to the centerline of the recessed channel. In some implementations, the floor is inwardly pointed or has a V shape that faces outwardly.

[0030] In some implementations, at least some of the recessed channels comprises a secondary channel extending along a centerline of the recessed channel, wherein the secondary channel provides the floor or a secondary floor recessed from the floor. In some implementations, the secondary channel has a uniform depth, a uniform width or both a uniform depth and a uniform width as the secondary channel extends between the first pole and the second pole within the recessed channel. In some implementations, the floor or secondary floor provided by the secondary channel is concave in a direction perpendicular to the centerline of the secondary channel and is convex in a direction along the centerline of the secondary channel. In some implementations, the floor or secondary floor provided by the secondary channel is flat or convex in a direction perpendicular to the centerline of the secondary channel and is convex in a direction along the centerline of the secondary channel.

[0031] In some implementations, the transitioning sides are concave in a direction from the floor to opposite edges of the recessed channel. In some implementations, the transitioning sides linearly extend from the floor to the opposite edges of the recessed channel. In some implementations, the transitioning sides each have a first portion extending from the floor towards side edges of the recessed channel that is concave and a second portion extending from the side edges of the recessed channel towards a first portion that is convex.

[0032] In some implementations, the transitioning sides of the recessed channels have a first depth at a midpoint between the first pole and the second pole, and have a second depth, less than the first depth, at or proximate the first pole and at or proximate the second pole. In some implementations, the transitioning sides have a first width at a midpoint between the first pole and the second pole and have a second width, less than the first width, at or proximate the first pole and at or proximate the second pole. In such implementations, the floors extending between the transitioning sides may have a constant or uniform width or may have a width that varies.

[0033] In some implementations, the game ball (excluding the recessed channels) is spherical, such as having the shape of a basketball. In such implementations, the game ball may comprise eight lobes extending between the first pole and the second pole. In some implementations, game ball may be in the form of a prolate spheroid, wherein the first pole and the second pole are at the tips of the prolate spheroid. Examples of such a prolate spheroid include an American football or rugby

ball. In such implementations, the prolate spheroid may comprise multiple lobes formed by individual panels or having the appearance of individual panels (such as in the case of a molded or 3D printed prolate spheroid), wherein the real or aesthetic panels have recessed seams that extend between the poles and that form the recessed channels, wherein the recessed channels vary in at least one of width, depth or cross-sectional shape to provide enhanced gripping of the prolate spheroid.

[0034] In some implementations, the recessed channels vary in at least one of width, depth and cross-sectional shape. In other implementations, some of the recessed channels may be uniform in width, depth and cross-sectional shape as they extend between the first pole and the second pole. One or more of the other recessed channels may vary in width, depth and/or cross-sectional shape as they extend between the first pole and the second pole. In some implementations, an individual recessed channel may vary along its length in at least one of width, depth and cross-sectional shape. In some implementations, a first one of the recessed channels may be uniform along its length and a second one of the recessed channels may also be uniform along its length, but wherein the first one of the recessed channels is different than the second one of the recessed channels in at least one of width, depth and cross-sectional shape at at least one location along its length. In some implementations, a first one of the recessed channels may vary along its length in at least one of width, depth and cross-sectional shape and a second one of the recessed channels may also vary along its length in at least one of width, depth and cross-sectional shape, where the first one of the recessed channels is also different than the second one of the recessed channels in at least one of width, depth and cross-sectional shape in corresponding locations along their lengths.

[0035] In some implementations, the varying recessed channels are formed during molding or 3D printing of the outer surface. In the case of molding, the mold forms the recessed channels that vary. In some implementations, the game ball comprises an underlying carcass that is molded, 3D printed or shaped so as to form the varying recessed channels, wherein outer cover panels of uniform thickness are placed over the carcass and cooperate with the underlying carcass to form the varying recessed channels. In some implementations, the game ball comprises an underlying carcass that does not form the varying recessed channels, wherein outer cover panels have a nonuniform thickness, where the cover panels have tapered edges that vary and are placed over the carcass to form the varying recessed channels. In some implementations, the recessed channels separate and space opposite edges of the cover panels. In some implementations, the edges of the cover panels are in abutment, wherein the edges of the cover panels form the varying recessed channels.

[0036] FIGS. 1 and 2 are sectional views illustrating portions of an example variable loby game ball 20. Game ball 20 is a “loby” game ball in that it is a lobular, having an outer surface comprising multiple lobes. For purposes of this disclosure, the term “lobular” and “loby” refer to the surface of a region having a smaller radius of curvature as compared to the overall radius of the ball surface. Game ball 20 comprises a pair of opposite poles 22, 24 joined by a longitudinal axis 26. In the example illustrated, FIG. 1 is a sectional view taken along a first radial plane perpendicular to axis 26 at an equator or midpoint of the game ball 20 between the opposite poles 22, 24. FIG. 2 is a sectional view taken along a second plane perpendicular to axis 26 at a location along axis 26 between the first plane and one of poles 22, 24.

[0037] As shown by FIGS. 1 and 2, the outer surface of game ball 20 comprises multiple outwardly facing circumferential lobes 30-1, 30-2, 30-3, 30-4, 30-5, 30-6, 30-7 and 30-8 (collectively referred to as lobes 30) that extend along axis 26 between the opposite poles 22, 24. The outer surface of game ball 20 additionally comprises recessed channels 32-1, 32-2, 32-3, 32-4, 32-5, 32-6, 32-7, and 32-8 (collectively referred to as recessed channels 32). Lobes 30 comprise outer surface protuberances or circumferential portions circumferentially extend about axis 26. Although not illustrated, each of such surface portions may be pebbled or provided with other surface textures for enhanced gripping of the game ball 20.

[0038] Recessed channels 32 each extend along consecutive lobes 30 along axis 26 between poles 22, 24. In some implementations, recessed channels 32 extend, in an uninterrupted fashion from pole 22 to the other pole 24. Each of recessed channels 32 comprises a floor 40 and a pair of transitioning sides 42. Floor 40 is that portion of the recessed channel 32 that is closest to axis 26. Transitioning sides 42 extend outwardly (away from axis 26) into conjunction with the outermost circumferential surfaces of the adjacent lobes 30. In the example illustrated, each floor 40 is outwardly concave in a direction perpendicular to the centerline or length of the associated recessed channel 32 and is convex in a direction along axis 26, along the length of the associated recessed channel 32.

[0039] As shown by FIG. 1, the various recessed channels 32 vary or differ from one another. Channels 32-1 and 32-5 are similar to one another. Channels 32-2, 32-4, 32-6 and 32-8 are similar to one another. Channels 32-3 and 32-7 are similar to one another. However, recessed channels 32-1 and 32-5 are different in at least one of width, depth and cross-sectional shape as compared to recessed channels 32-2, 32-4, 32-6 and 32-7 as well as recessed channels 32-3, 32-7. Likewise, recessed channels 32-2, 32-4, 32-6 and 32-8 are different in at least one of width, depth and cross-sectional shape as compared to channels 32-3 and 32-7. The variability amongst such channels 32 may provide a customized outer surface for enhanced gripping of the game ball 20.

[0040] FIG. 2 illustrates a different portion of the outer surface of game ball 20. As shown by FIG. 2, at the illustrated position closer to one of poles 22, 24, recessed channels 32-1, 32-2, 32-4, 32-5, 32-6 and 32-8 are both shallower and narrower in FIG. 2 as compared to the depth and width of the same channels at the location shown in FIG. 1. In the example illustrated, the width and depth of recessed channel 32-3 and 32-7 is uniform from the midpoint of game ball 20 shown in FIG. 1 to the pole proximate position shown in FIG. 2. In other implementations, the width and/or depth of recessed channels 32-3 and 32-7 may likewise vary along their lengths or along axis 26.

[0041] Although game ball 20 is illustrated as comprising eight uniformly spaced lobes 30 and eight intervening recessed channels 32, in other implementations, game ball 20 may include a fewer or greater of such lobes and a greater or fewer of such recessed channels 32. In some implementations, the lobes and channels may be nonuniformly or non-symmetrically spaced about axis 26. In other implementations, the relative or proportional widths and depths may be changed. For example, FIGS. 1 and 2 currently illustrate channels 32 with exaggerated sizes to better illustrate their respective depths, widths and cross-sectional shapes.

[0042] FIGS. 3 and 4 are sectional views illustrating portions of an example variable loby game ball 120. FIGS. 3 and 4 illustrate various example shapes for the recessed channels that extend along axis 26 and along consecutive circumferential portions or lobes. As with game ball 20, loby game ball 120 comprises a pair of opposite poles 22, 24 joined by a longitudinal axis 26. FIG. 3 is a sectional view taken along a first plane perpendicular to axis 26 at an equator or midpoint of the game ball 120 between the opposite poles 22, 24. FIG. 4 is a sectional view taken along a second plane perpendicular to axis 26 at a location along axis 26 between the first plane and one of poles 22, 24.

[0043] As shown by FIGS. 3 and 4, the outer surface of game ball 120 comprises multiple outwardly facing circumferential lobes 130-1, 130-2, 130-3, 130-4, 130-5, 130-6, 130-7 and 130-8 (collectively referred to as lobes 130) that extend along axis 26 between the opposite poles 22, 24. The outer surface of game ball 120 additionally comprises recessed channels 132-1, 132-2, 132-3, 132-4, 132-5, 132-6, 132-7, and 132-8 (collectively referred to as recessed channels 132). Lobes 130 comprise outer surface protuberances or circumferential portions circumferentially extend about axis 26. Although not illustrated, each of such surface portions may be pebbled or provided with other surface textures for enhanced gripping of the game ball 120.

[0044] In the example illustrated, recessed channels 132-1 and 132-5 have similar widths, depths and cross-sectional shapes in the plane shown by FIG. 3. Channels 132-1 and 132-5 each have an outwardly facing “V” shape with a floor 140-1 that is inwardly pointed or also has a V shape while

facing outwardly. The transitioning sides **142-1** of channel **32-1** outwardly diverge from the floor. In some implementations, channels **132-1**, **132-5** may be formed by stitching to opposing panels together along axis **26** and turning the panel inside out. In other implementations, channels **132-11** **32-5** may be 3D printed, molded or formed through material removal processes.

[0045] Recessed channels **132-2** and **132-6** have similar widths, depths and cross-sectional shapes in the plane shown by FIG. 3. Channels **132-2** and **132-6** each have a generally flat or planar floor **140-2** that is convex along axis **26**, and a pair of transitional sides **142-2** that are outwardly convex along axis **26** from the sides of floor **140** to the outer most portions of the adjacent lobes **130**.

[0046] Recessed channels **132-3** and **132-7** have similar widths, depths and cross-sectional shapes in the plane shown in FIG. 3. Channels **132-3** and **132-7** each have a generally flat or planar floor **140-3** and transitioning sides **142-3**. Transitioning sides **142-3** each have a first portion extending from the floor **140-3** towards side edges of the recessed channel **132-3**, **132-7** that is concave and a second portion extending from the side edges of the recessed channel **132-3**, **132-7** towards a first portion that is convex. As result, recessed channels **132-3**, **132-7** have an “S” shape, providing a smooth transition to the outermost surface of the adjacent lobes **130**.

[0047] As shown by FIG. 3, the various recessed channels **132** vary or differ from one another. Channels **132-1** and **132-5** are similar to one another. Channels **132-2**, and **132-6** are similar to one another. Channels **132-3** and **32-7** are similar to one another. Channels **132-4** and **132-8** are similar to one another. However, recessed channels **132-1** and **132-5** are different in at least one of width, depth and cross-sectional shape as compared to recessed channels **132-2** and **132-6** are different in at least one of width, depth and cross-sectional shape as compared to the other recessed channels of ball **120**. Likewise, recessed channels **132-3** and **132-7** are different in at least one of width, depth and cross-sectional shape as compared to the other channels of ball **120**. Recessed channels **132-4** and **132-8** are different in at least one of width, depth and cross-sectional shape as compared to other channels of ball **120**. The variability amongst such channels **32** may provide a customized outer surface for enhanced gripping of the game ball **20**.

[0048] FIG. 4 illustrates a different portion of the outer surface of game ball **120**. As shown by FIG. 2, at the illustrated position closer to one of poles **22**, **24**, each of the recessed channels **132** is shallower and narrower in FIG. 4 as compared to the depth and width of the same channels at the location shown in FIG. 3.

[0049] Although game ball **120** is illustrated as comprising eight uniformly spaced lobes **130** and eight intervening recessed channels **132**, in other implementations, game ball **120** may include a fewer or greater of such lobes and a greater or fewer of such recessed channels **132**. In some implementations, the lobes and channels may be nonuniformly or non-symmetrically spaced about axis **26**. In some implementations, game ball **120** may have a larger variety of differently shaped recessed channels or a lesser variety of differently shaped recessed channels. In some imitations, every recessed channel of game ball **120** may have one of the four different shapes shown in FIG. 3, wherein such recessed channels vary in at least one of width, depth or cross-sectional shape along their individual lengths along axis **26**. In other implementations, the relative or proportional widths and depths may be changed. For example, FIGS. 3 and 4 currently illustrate channels **132** with exaggerated sizes to better illustrate their respective depths, widths and cross-sectional shapes.

[0050] The lobes and intervening recessed channels shown in FIGS. 1-4 may be used in a variety of different game balls. For example, FIG. 5 illustrates an example game ball **220** in the form of a prolate spheroid, sometimes referred to as an American football or rugby ball. The outer surface of game ball **20** comprises poles **222** and **224** and comprises four lobes **230-1**, **230-2** and two additional opposite lobes (not shown) extending between poles **222** to **224**. Recessed channels, such as the recessed channels **140-1** described above, may be provided along consecutive or adjacent lobes **230**. The recessed channels **140-1** may vary in depth and width along their length. For example, portions **244** may be provided with a recessed channel **140-1** having a first width and a first depth (similar to shown in FIG. 3) and portions **245** (proximate to poles **222** and **224**) are

provided with the recessed channel **140-1** having a second width and a second depth (similar to shown in FIG. **4**), the second width being narrower than the first width and the second depth being shallower than the first depth. The wider and deeper portion of recessed channel **140-1** proximate to the laces **246** may provide enhanced gripping of the prolate spheroid. In other implementations, recessed channels of the game ball **220** may have other shapes and variations as described above. [0051] FIG. **6** illustrates an example spherical lobby game ball **260** in the form of a basketball. Game ball **260** comprises poles **262** and **264**, lobes **270** and recessed channels **272**. As with lobes **30** and **130** described above, lobes **270** comprise outer surface protuberances or circumferential portions circumferentially extend about axis **266**. Although not illustrated, each of such surface portions may be pebbled or provided with other surface textures for enhanced gripping of the game ball **260**.

[0052] Recessed channels **272** each extend along consecutive lobes **270** along axis **266** between poles **222**, **224**. Each of channels **272** may be similar to any of recessed channels **32** and **132** described above. For example, in some implementations, each of channels **272** may have the shape of any one of channels **32** or **132** described above, wherein all the individual channels **272** are identical to one another along their lengths (the same cross-sectional shape, depth and width at corresponding locations along their lengths) but for their potentially differing paths along the surface of game ball **260**. In such implementations, each of such channels **272** may vary along its length in at least one of width, depth and cross-sectional shape as described above with respect to channels **32** and **132** (for example, similar to how the width and depth of channel **132-8** changes from the midpoint shown in FIG. **3** to the pole proximate location shown in FIG. **4**).

[0053] In some implementations, different individual channels **272** may have a uniform width, depth and cross-sectional shape along their individual lengths, but where the individual channels **272** differ amongst themselves in at least one of width, depth and cross-sectional shape at corresponding locations (for example, similar to how the width, depth and shape of channel **130-7** is different from the width, depth and shape of channel **132-8** in FIG. **3**). In some implementations, a first one of channels **272** may have a first width, a first depth and a first cross-sectional shape at an equator of the game ball **260** and a second one of the channel **272** has a second width, a second depth and a second cross-sectional shape at the equator of the game ball **260**, wherein at least one of the second width, the second depth or the second cross-sectional shape differs from the first width, the first depth or the first cross-sectional shape, respectively. In some implementations, recessed channels **272** may both vary or differ amongst themselves in at least one of a width profile, a depth profile and a cross-sectional shape profile and may also individually vary or differ along their individual lengths in at least one of width, depth and cross-sectional shape. In some implementations, portions of a first individual channel **272** may be similar to corresponding portions of a second individual channel **272** while other portions of the first individual channel **272** may be different from corresponding other portions of the second individual channel **272**.

[0054] FIGS. **7-10** illustrate an example spherical game ball **320** in the form of a basketball. Game ball **320** has an outer surface comprising poles **322** and **324** located along a longitudinal axis **326**. Game ball **320** further comprises multiple outwardly facing circumferential lobes **330-1**, **330-2**, **330-3**, **330-4**, **330-5**, **330-6**, **330-7** and **330-8** (collectively referred to as lobes **330**) that extend along axis **326** between the opposite poles **322**, **324**. The outer surface of game ball **320** additionally comprises recessed channels **332-1**, **332-2**, **332-3**, **332-4**, **332-5**, **332-6**, **332-7**, and **332-8** (collectively referred to as recessed channels **332**). Lobes **330** comprise outer surface protuberances or circumferential portions that circumferentially extend about axis **326**. Although not illustrated, each of such surface portions may be pebbled or provided with other surface textures for enhanced gripping of the game ball **320**.

[0055] Recessed channels each extend along consecutive lobes **330** along axis **26** between poles **322**, **324**. Recessed channels **332-1** and **332-5** are coplanar, extending directly opposite to one another and extending from pole **322** to pole **324**. Likewise, recessed channels **332-3** and **332-7** are

coplanar, extending directly opposite to one another and extending from pole **322** to pole **324**. Recessed channels **332-3** and **332-7** are angularly offset from channels **332-1** and **332-5** by 90 degrees about axis **326**.

[0056] Recessed channel **332-2** extends between lobes **330-1** and **330-2**, equidistantly spaced from channels **332-1** and **332-3** at a midpoint or equator of game ball **320**. Similarly, recessed channel **332-8** extends between lobes **330-7** and **330-7**, equidistantly spaced from channels **332-1** and **332-7** at a midpoint or equator of game ball **320**. Channels **332-2** and **332-8** do not extend completely to poles **322** and **324**. Rather, channels **332-2** and **332-8** have opposite end portions **333** that turn towards and converge with channel **332-1** at locations in close proximity to poles **322** and **324**. End portions **333** of channels **332-2** and **332-8** are aligned with one another on opposite sides of channel **332-1**.

[0057] Recessed channel **332-4** extends between lobes **330-3** and **330-4**, equidistantly spaced from channels **332-3** and **332-5** at a midpoint or equator of game ball **320**. Similarly, recessed channel **332-6** extends between lobes **330-5** and **330-6**, equidistantly spaced from channels **332-5** and **332-7** at a midpoint or equator of game ball **320**. Channels **332-4** and **332-6** do not extend completely to poles **322** and **324**. Rather, channels **332-4** and **332-8** have opposite end portions **335** that turn towards and converge with channel **332-5** at locations in close proximity to poles **322** and **324**. End portions **335** of channels **332-4** and **332-6** are aligned with one another on opposite sides of channel **332-5**.

[0058] Each of recessed channels **332** comprises a secondary channel **338** having a floor **340**, and a pair of transitioning sides **342**. As seen in FIGS. **14** and **16**, secondary channel **338** may be further recessed from transitioning sides **342**. Floor **340** may serve as the floor of each of channels **332**. In the example illustrated, floor **340** is outwardly concave (inwardly curved) in a direction perpendicular to axis **326** while being convex in a direction along axis **326**. In other implementations, floor **340** may be flat in a direction perpendicular to axis **326** while being convex in a direction along axis **326**. In still other implementations, floor **340** may be convex in a direction perpendicular to axis **326** and convex in a direction along axis **326**. In some implementations, secondary channel **338** may not be recessed from transitioning sides **342**, wherein sides of floor **340** merge or converge with the lower edges of transitioning sides **342**.

[0059] As shown by FIGS. **7-10**, the various recessed channels **332** are each similar to one another but for the paths that they take along the surface of game ball **320**. Each of channels **332** varies in at least one of depth, width and length along its length as it extends between poles **322** and **324**. In the example illustrated, secondary channel **338** has a uniform width and depth along its length. It is the recessed transitioning sides **342** that vary along the length of each of channels **332** such that channels **332** vary along their lengths. In the example illustrated, each of the pair of transitioning sides on opposite sides of the second channel **338** are similar to one another, mirroring one another. In other implementations, secondary channel **338** may also vary in at least one of width, depth and cross-sectional shape along the length of the channel **332**. In other implementations, the two transitioning sides on opposite sides of the same secondary channel **338** may differ from one another in at least one of width, length and cross-sectional shape.

[0060] As shown by FIGS. **11** and **12**, the width of each of recessed channels **332** varies along its length. As shown by FIG. **11**, at the plane **367** taken through the midpoint or equator of game ball **320**, between poles **322**, **324**, each of the transitioning sides **342** on opposite sides of the floor **340** of the secondary channel **338** have a width **W1** (also shown in FIGS. **9** and **10**) in a direction perpendicular to the longitudinal axis **326** of game ball **320**. As shown by FIG. **12**, the same channel has a width **W2** (also shown in FIGS. **9** and **10**) in a direction perpendicular to the longitudinal axis **366** proximate to each of poles **322**, **324**. The width **W-2** is smaller than the width **W1**. In some implementations, width **W** is at least 3 mm wider than width **W-2**. As a result, the overall width of each of channel **332** at its midpoint is at least 6 mm wider than the overall width of the same channel **332** proximate to poles **322** and **324**. In some implementations, width **W1** is at

least 7 mm and no greater than 10 mm, and in some implementations, 8.36 mm. In such implementations, width W2 is at least 4 mm and no greater than 7 mm, and in some implementations, 5.33 mm.

[0061] As shown by FIGS. **13-16**, the depth of each of the transitioning sides **342** of recessed channels **332** varies along its length. As shown by FIG. **14**, at the plane **367** taken through the midpoint or equator of game ball **320**, between poles **322**, **324**, each of the transitioning sides **342** on opposite sides of the floor **340** of the secondary channel **338** have a depth D1. As shown by FIG. **16**, the same transitioning side **342** has a depth D2. Depth D1 and depth D2 preferably are sized to be within the range of 1.5 to 5.0 mm. In some implementations, depth D2 may be smaller than depth D1. In other implementations, the depths D1 and D2 can be substantially the same. The greater depth at the midpoint or a more central portions of each of channels **332** may provide enhanced feel and enhance gripping of game ball **320**. The enhance gripping may facilitate the generation of greater backspin during shooting of the game ball **320**. In the example illustrated, the change in depth from the midpoint to the poles is gradual. In other implementations, the change in depth may occur in a stepwise fashion.

[0062] As further shown by FIGS. **14** and **16**, the cross-sectional shape of each of channels **332** changes along its length. As shown by FIGS. **13** and **14**, at the midpoint or at points in a plane taken through the equator of game ball **320**, each of the transitioning sides **342** of each of channels **332** has an outwardly concave shape (in a direction perpendicular to the longitudinal axis **366**) that changes to an outwardly convex shape prior to converging with the adjacent lobes **330**. As shown by FIGS. **15** and **16** proximate to the poles **322**, **324**, each of the transitioning sides **342** of each of the channels **332** is in the shape of a linear ramp transversely extending from the circumferential surfaces of the adjacent lobes **330** to the outer sides or edges of the recessed secondary channel **338**.

[0063] In the example illustrated, despite the changes in depth and width along the length of each of channels **332**, transitioning from a wider and deeper sides **342** at and near the equator of the game ball to a narrower and shallower sides **342** proximate the poles of the game ball, such channels **332** are configured so as to provide a largely consistent angle along each length. The unique variable lobular shape or loby shape of the game ball of the implementations of the present invention provide additional grip points to the central regions of the game ball **320** between the poles **322** and **324**. The pole regions of the game ball **320** and most game balls have high number of grip points for a player. The central region of the game ball **320** between the poles **322** and **324** offer additional grip points that facilitate a player's ability to grasp, shoot, pass, dribble and otherwise handle the game ball **320**. The central region is also the preferred region of player's to hold or contact the ball when shooting, time permitting.

[0064] FIGS. **17**, **18** and **19** illustrate an individual channel **332** at the shallowest location along the channel (proximate to one of poles **322**, **324**) for its sides **342**, at the beginning of a channel drop transition along the channel, and at a deepest side depth along the channel (portions of the channel extending across the game ball equator), respectively. As shown by as shown by FIG. **17**, transitioning sides **342** each have an angle A1 at the shallowest location on along the channel **332**. As shown by FIG. **18**, transitioning sides **342** each have an angle A2 at locations where the channel side depth begins to deepen. As shown by FIG. **19**, at the deepest locations of channel sides **342**, each of transitioning sides **342** have an angle A3. Angles A1, A2 and A3 are all within a range of 10 degrees, and in the example illustrated, within a range of 5 degrees. In the example illustrated, angle A1 is approximately 83.95°; angle A2 is approximately 83.99° and angle A3 is approximately 80.99 degrees. In other implementations, the angles of the transitioning sides **342** (the lobiness angle) may be less consistent along the length of each of channels **332**. In some implementations, some sides **342** may have more consistent angles along their lengths than others.

[0065] FIGS. **20-22** illustrate example game balls **420**, **520** and **620**, respectively, in the form of basketballs. Game ball **420** is similar to game ball **320** except that each of its channels **332** are

wider at the equator than the channels **332** of game ball **320**. Game ball **520** is similar to game ball **420** except that each of its channels **332** are even wider still at the equator than the channels **332** of game ball **420**. Game ball **620** is similar to game ball **520** except that each of its channels **332** are still wider at the equator than the channels **332** of game ball **520**. As shown by FIG. 20-22, the width, depth and cross-sectional shape of channels **332** may be varied along their lengths in different manners to accommodate different playing environments.

[0066] As with the basketball shown in FIGS. 9-19, each of the basketballs shown in FIGS. 20-22 have a size, weight, and rebound consistency similar to that of standard or conventional competitive play basketballs currently sanctioned by various organizations such as the National basketball Association (NBA), National Collegiate Athletic Association (NCAA), National Federation of High Schools (NFHS) and other organizations. Each of game balls **320**, **420**, **520** and **620**, when dropped from a height of 72" measured from the bottom of the ball, has a minimum rebound of 49" for a retail ball (measured from top of ball) and 52" for an NBA game ball. Each of such game ball **320**, **420**, **520** and **620** has a maximum rebound of 56 inches for an NBA ball.

[0067] FIGS. 23-25 illustrate various example game ball constructions which illustrate different ways by which the varying recess channels described above may be formed. FIGS. 23-25 illustrate different example constructions for game balls **320**, **420**, **520** and **620** described above. FIG. 23 is a sectional view through a portion of an example game ball **720**, which may be in the form of a basketball. Game ball **720** may have an outer surface similar to the above-described outer surface of game ball **320**, wherein game ball **720** comprises each of the above-described lobes **330** and recessed channels **332**. Conversely, in some implementations, game ball **320** may have the cross-section shown in FIG. 23 along each of its channels **332** and may be formed as described below with respect to game ball **720**.

[0068] Game ball **720** comprises inflatable bladder **828**, windings **830**, outer layer **834** and cover panels **835**. In the example illustrated, bladder **828** may be formed primarily of rubber or a synthetic rubber. In one implementation, bladder **828** may be formed from an 80% butyl rubber and 20% natural rubber. During manufacture, the bladder may be inflated and placed in a vulcanizing or curing mold where the bladder is cured at a temperature of 160° C. After curing, the inflated bladder may be wound with a reinforcing thread to form the layer of windings **830**. The layer windings may be formed from a polymer or other materials. In the example illustrated, layer of windings **830** may be formed from the approximately 2100 meters of 210 denier Nylon 66 thread. The thread may be coated with a latex or adhesive.

[0069] Outer layer **834** comprises a layer of rubber and/or synthetic rubber formed over windings **830**. In some implementations, layer **834** may be formed by flat sheets or panels of rubber, such as natural and/or butyl rubber or sponge rubber (see U.S. Pat. No. 5,681,233, hereby incorporated by reference), which are laid over the wound bladder (layers **828** and **830**) so as to completely cover the wound bladder. In such implementations, the inflated rubber-covered bladder is in place in a spherical carcass mold and which may be heat molded at about 160° C. to form the carcass formed by layers **828**, **830** and **834**. During such molding, the rubber panels fuse to each other and into the windings **830** to form an integral outer layer **834** on the carcass.

[0070] The inside surface of the carcass mold may be shaped to form the contours of the lobular outer outside surface of game ball **720**. In particular, the inside surface of the carcass mold may be shaped or configured to form each of the foundational lobes **730** and each of the recessed channels **732** in layer **834**. Each of lobes **730** and recessed channels **732** may be similar to lobes **330** and channels **332**, respectively, as described above and may have the layout about the game ball as shown above with respect to game ball **320**. As shown by FIG. 23, each of the recessed channels formed in layer **834** comprises a secondary channel **738** having a floor **740** and a pair of transitioning sides **742** on opposite sides of floor **740**. Secondary channel **738**, floor **740** and sides **742** may be similar or identical to channels **338**, floors **340** and sides **342** described above with respect to any of game balls **320**, **420**, **520** and **620**. As with sides **342**, sides **742** vary in depth,

with cross-sectional shape along the length of each of channels **732**. As with sides **342**, sides **742** may be deeper and wider at an equator of the game ball and shallower and narrower proximate to the poles of the game ball.

[0071] Cover panels **835** extend over layer **834**. Each of cover panels has a uniform thickness such that the outer surface contour of layer **834** is maintained. The outer surface of the completed game ball **720** has the same shape as the outer surface of layer **834**. The outer surface of the completed game ball **720** comprises lobes **330**, recessed channels **332**, and secondary channels **338** (deeper than secondary channel **738**) having the floor **740** which corresponds to floor **340**. In some implementations those portions of cover panels **835** overlying sides **742** may be thinner or skived (trimmed cut away at an angle across the corner of the edges) or otherwise tapered to modify the shape, depth or width of the outermost recessed channels **332**.

[0072] In some implementations, cover panels **835** may be formed from materials such as leather, synthetic leather, rubber and the like. The outer surface of each cover panel **835** may provide with a pebbled texture. Each of cover panels **835** may have a shape corresponding to the surface area of carcass which is bounded by secondary channels **738**. Each of such cover panels may be adhered to layer **834** within adhesive, wherein portions of layer **34** may likewise be coated with an adhesive. In some implementations, each of cover panels **835** may have a uniform thickness of 0.031 inches (0.80 mm) throughout the panel except the variation in thickness caused by the pebbled outer surface. In other implementations, cover panel **835** may have different thicknesses within the range of 0.721.25 millimeters. In some implementations, cover panels **835** may be omitted, wherein the outer surface of layer **834** forms the outer surface of the completed game ball **720**. In such implementations, the outer surface of layer **834** may be pebbled. In some implementations, windings **830** may be omitted.

[0073] FIG. **24** illustrates game ball **920**, another example game ball in the form of a basketball. Game ball **920** is similar to game ball **720** except that ball **920** comprises outer layer **934** in place of outer layer **834**. Those remaining components of game ball **920** which correspond to components of game ball **720** (and game ball **320**) are numbered similarly.

[0074] Outer layer **934** is similar to outer layer **834** except that outer layer **934** comprises an elevated secondary channel **938** in place of the recessed secondary channel **738** in each of the channels **732**. The elevated secondary channel **938** comprises a wall **939** protruding above the deepest portions of sides **742** and having a top with a floor **940**. In the example illustrated, floor **940** is concave in a direction perpendicular to the longitudinal axis, such as axis **366** of the game ball and convex in a direction along the longitudinal axis of the game ball. In the example illustrated, the maximum depth of floor **940** remains above the deepest portions of sides **742**. In other implementations, floor **940** may be flat in directions perpendicular to the longitudinal axis and convex in directions along the longitudinal axis. In other implementations, floor **940** may be rounded or convex in directions perpendicular to the longitudinal axis and convex in directions along the longitudinal axis. In some implementations, floor **940** may be textured. For example, floor **940** may be pebbled.

[0075] As with game ball **720**, because cover panels **835** have a uniform thickness, the outer surface of the completed game ball **920** between elevated secondary channels **938** has the same shape as the outer surface of layer **934**. The depth and width of the transitioning sides **742** formed in layer **934** are maintained to form transitioning sides **342** which extend between lobes **330**. In the example illustrated, the thickness of cover panels **835** is less than or equal to the height of elevated channels **938**; the thickness of cover panels **835** is less than or equal to the distance that elevated channels **938** protrude or rise above the deepest portions of sides **742**.

[0076] In other implementations, cover panels **835** may have a greater thickness such that the outer surface of cover panels **835** extends above elevated channels **938**. In such implementations, edge portions of cover panels **835** extending along an adjacent to elevated channel **938** may be tapered, skived or otherwise thinner. In some implementations, cover panels **835** may be omitted from game

ball **920**. In such implementations, the outer surface of layer **934** forming lobes **730**, sides **742** and floor **940** may be pebbled for enhanced grip. In some implementations, as shown by broken lines **941**, rather than being integrally formed as a single unitary body as part of layer **934**, the elevated channel **938** may comprise a separate strip of material that fused or bonded to a top of layer **934** or that is inset into a groove formed in layer **934** prior to being fused or bonded in place.

[0077] FIG. **25** illustrates game ball **1020**, another example game ball in the form of a basketball. Game ball **1020** is similar to game ball **720** except that ball **920** comprises outer layer **1034** and cover panels **1035** in place of outer layer **834** and cover panels **835**, respectively. Those remaining components of game ball **920** which correspond to components of game ball **720** (and game ball **320**) are numbered similarly.

[0078] Outer layer **1034** is similar to outer layer **834** except that outer layer **1034** does not include sides corresponding to sides **342** of the recessed channels **332**. In contrast, outer layer **1034** may have a uniform thickness about the entire circumference of the game ball on **1020**. Portions of the outer layer **1034** form floors **340** of each of the recessed channels **332**. Cover panels **1035** are similar to cover panels **835** described above except that edge portions **1036** of cover panel **1035** are shaped, thinned, tapered, cut-away or skived so as to form the above-described transitioning sides **342** along the opposite sides of each of above-described channels **332**.

[0079] In some implementations, those portions of layer **1034** serving as a floors **340** of the channels **332** may have a different thickness than remaining portions of the layer **1034**. For example, such portions may be concave in directions perpendicular to the longitudinal axis of game ball or convex in directions perpendicular to the longitudinal axis of the game ball (transversely across channel **332**). Such portions may be pebbled. As described above with respect to FIG. **24**, in some implementations, layer **1034** may include an integrally formed elevated channel **938** between the transitioning sides **342** provided by the edges of panel **1035**. In some implementations, as shown by broken lines **941** in FIG. **24**, and elevated channel **938** in the form of strip or multiple strips may be fused or bonded to a top of layer **1034** or inset within a groove or channel formed within layer **1034** between the tapered, cut-away or skived edges of panel **1035**.

[0080] Although the present disclosure has been described with reference to example implementations, workers skilled in the art will recognize that changes may be made in form and detail without departing from the scope of the claimed subject matter. For example, although different example implementations may have been described as including features providing benefits, it is contemplated that the described features may be interchanged with one another or alternatively be combined with one another in the described example implementations or in other alternative implementations. Because the technology of the present disclosure is relatively complex, not all changes in the technology are foreseeable. The present disclosure described with reference to the example implementations and set forth in the following claims is manifestly intended to be as broad as possible. For example, unless specifically otherwise noted, the claims reciting a single particular element also encompass a plurality of such particular elements. The terms “first”, “second”, “third” and so on in the claims merely distinguish different elements and, unless otherwise stated, are not to be specifically associated with a particular order or particular numbering of elements in the disclosure.

Claims

1. A variable loby game ball comprising: an outer surface comprising: a first pole and a second pole opposite the first pole; outer lobes extending between the first pole and the second pole, herein recessed channels that vary in at least one of width, depth and cross-sectional shape extend along consecutive lobes.

2. The variable loby game ball of claim 1, wherein the recessed channels each comprise a floor and a pair of transitioning sides, wherein the transitioning sides vary in at least one of width, depth and

cross-sectional shape.

3. The variable loby game ball of claim 2 wherein each of the recessed channels has a secondary channel forming the floor between the pair of transitioning sides.
 4. The variable loby game ball of claim 2, wherein the transitioning sides extend from proximate the first pole to proximate the second pole and wherein the transitioning sides vary in at least one of depth and width along their lengths between the first pole and the second pole.
 5. The variable loby game ball of claim 4, wherein the transitioning sides vary in both depth and width along their lengths between the first pole and the second pole.
 6. The variable loby game ball of claim 4, wherein the transitioning sides vary in width along their lengths between the first pole and the second pole.
 7. The variable loby game ball of claim 6, wherein the transitioning sides have a first width proximate the first pole and the second pole and a second width, greater than the first width, at a midpoint between the first pole and the second pole.
 8. The variable loby game ball of claim 7, wherein the transitioning sides have a first depth proximate the first pole and the second pole and a second depth, greater than the first depth, at the midpoint between the first pole and the second pole.
 9. The variable loby game ball of claim 8, wherein the pairs of transitioning sides proximate the first pole and the second pole extend at a first angle towards the channel and wherein the pairs of transitioning sides at the midpoint between the first pole and the second pole extend towards the channel at a second angle, the second angle being within 5 degrees of the first angle.
 10. The variable loby game ball of claim 7, wherein the transitioning sides are concave at the midpoint and convex proximate the first pole and the second pole.
 11. The variable loby game ball of claim 7, wherein the transitioning sides extend an angle of at least 75 degrees and no greater than 88 degrees relative to a radial plane passing through a center of the variable loby game ball.
 12. The variable loby game ball of claim 4, wherein the transitioning sides extending along the opposite sides of each channel and mirror one another.
 13. The variable loby game ball of claim 2, wherein the recessed channels comprise a first channel having a first pair of transitioning sides and a second channel having a second pair of transitioning sides different with respect to at least one of width, depth and cross-sectional shape of the first pair of transitioning sides.
 14. The variable loby game ball of claim 13, wherein the transitioning sides extend from proximate the first pole to proximate the second pole and wherein the transitioning sides vary in at least one of depth and width along their lengths between the first pole and the second pole.
 15. The variable loby game ball of claim 14, wherein the transitioning sides vary in both depth and width along their lengths between the first pole and the second pole.
 16. The variable loby game ball of claim 14, wherein the transitioning sides have a first width proximate the first pole and the second pole and a second width, greater than the first width, at a midpoint between the first pole and the second pole.
 17. The variable loby game ball of claim 16, wherein the transitioning sides are concave at the midpoint and convex proximate the first pole and the second pole.
 18. The variable loby game ball of claim 2 further comprising: a carcass; and a plurality of cover panels secured to the carcass.
 19. The variable loby game ball of claim 18, wherein the carcass has a lobular outer carcass surface having foundational lobes corresponding to the lobes of the outer surface.
 20. The variable loby game ball of claim 19, wherein the plurality of cover panels have a non-uniform thickness so as to form the pair of transitioning sides along the floor.
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